Filter Functions for Reconstruction Algorithms

Kevin Velez and Juan Lopez

Abstract The Filtered Backprojection (FBP) is a reconstruction algorithm that uses pseudo-differential operators as filters. In the present extended abstract we explore the FBP reconstruction with multiple pseudo-differential filters, giving some results that explain the behavior of the numerical simulations. With this in mind, we propose an alternative perspective aimed at optimizing the discrete FBP.

1 Introduction

The Radon transform, introduced by J. Radon in 1917 [1], associates to a function f defined on $\mathbb{R}^n$ its integrals over affine hyperplanes. More precisely, for $f \in C_c^\infty(\mathbb{R}^n)$, the Radon transform $\mathcal{R}f$ is defined on $\mathbb{S}^{n-1} \times \mathbb{R}$ by integrating f over the hyperplane $\{x \in \mathbb{R}^n : x \cdot \theta = s\}$. From a functional analytic viewpoint, $\mathcal{R}$ can be regarded as a linear operator mapping functions on $\mathbb{R}^n$ into functions on the parameter space of hyperplanes. The central problem consists in recovering f from $\mathcal{R}f$, which constitutes a prototypical example of an inverse problem. Beyond its well-known applications in computerized tomography [3, 4], the Radon transform plays a fundamental role in harmonic analysis and integral geometry, where its inversion and regularity properties are closely connected to Fourier analysis and pseudo-differential operator theory [2].

The interception with the pseudo-differential operators occurs with the *Filtered Back-Projection* algorithms, where the adjoint operator of the Radon transform acts as a back-projection, and a pseudo-differential filter is applied. The fact that the filter operator is a pseudo-differential operator allows us to study the reconstruction with

Kevin Velez
Universidad del Valle, Cali, Colombia, e-mail: velez.kevin@correounivalle.edu.co

Juan Lopez
Universidad del Valle, Cali, Colombia, e-mail: juan.lopez.holguin@correounivalle.edu.co

a function defined in the Fourier domain, this function is called the *symbol* of the filter operator.

As a consequence, in order to obtain the properties of the filter we just need to look at its symbol. Equivalently, we can propose different symbols to obtain filters with different actions. This relation has been used recently in [10], where a low-pass filter that minimizes the expected squared error given infinite noisy measurement data is obtained.

In Section 2 we present some of the results in the theory of pseudo-differential operators that applies to Fourier multipliers. The discretization process is described in Section 3 and the conclusions are presented in Section 4.

2 Fourier Multipliers

The filtered back-projection (FBP) formula in two dimensions can be written as

$$f(x,y) = \frac{1}{2}\mathcal{R}^*\Big(\mathcal{F}^{-1}[\,|\sigma|\,\mathcal{F}(\mathcal{R}f)(\sigma,\varphi)]\Big)(x,y), \qquad (x,y)\in\mathbb{R}^2,$$

where $\mathcal{F}$ denotes the one-dimensional Fourier transform in the radial variable and $\mathcal{R}^*$ is the *backprojection* operator.

The operator $\mathcal{R}$ acts on integrable functions with compact support $f:\mathbb{R}^2\to\mathbb{R}$ as

$$\mathcal{R}f(t,\theta) = \int_{\{x\cos(\theta)+y\sin(\theta)=t\}} f(x,y)\mathrm{d}x\,\mathrm{d}y, \tag{1}$$

for $(t,\theta)\in\mathbb{R}\times[0,\pi)$. Meanwhile, the backprojection operator is defined as

$$\mathcal{B}h(x,y) = \frac{1}{\pi}\int_0^\pi h(x\cos(\theta)+y\sin(\theta),\theta)\mathrm{d}\theta, \tag{2}$$

for $(x,y)\in\mathbb{R}^2$ [9].

The filtration step

$$h \longmapsto \mathcal{F}^{-1}[\,|\sigma|\,\mathcal{F}h\,]$$

is a Fourier multiplier operator; more generally, given a measurable symbol $\tau:\mathbb{R}\to\mathbb{C}$, one defines

$$(T_\tau h)(s,\varphi) := \mathcal{F}^{-1}\big(\tau(\sigma)\,\mathcal{F}[h](\sigma,\varphi)\big)(s,\varphi).$$

Fourier multipliers constitute a basic subclass of pseudo-differential operators. More precisely, a pseudo-differential operator P on $\mathbb{R}^n$ can be written in the form

$$Pf(x) = \frac{1}{(2\pi)^n}\int_{\mathbb{R}^n} e^{ix\cdot\xi}\,a(x,\xi)\,\widehat{f}(\xi)\,d\xi,$$

where $\widehat{f}$ denotes the n-dimensional Fourier transform on $\mathbb{R}^n$, the function $a(x, \xi)$ is called the symbol of P, and a satisfies the standard symbol estimates of some order $m \in \mathbb{R}$ [5]. In the special case where a is independent of x, the operator P reduces to a Fourier multiplier with symbol $a(\xi)$. Consequently, for each fixed φ, the filtration operator T_τ is a pseudo-differential operator in the radial variable s whose symbol is precisely $\tau(\sigma)$; in particular, the regularizing or amplifying effect of the filter is encoded by the order and smoothness of τ.

The classical ramp filter corresponds to the choice $\tau(\sigma) = |\sigma|$, which is a symbol of order 1 in the sense that it grows linearly as $|\sigma| \to \infty$. Consequently, T_τ acts as a differentiating operator in the radial variable, enhancing high-frequency components of the data. This explains the well-known instability of the unregularized FBP method under measurement noise, since the factor $|\sigma|$ amplifies the high-frequency part of the sinogram. A standard way to stabilize the reconstruction is therefore to replace $|\sigma|$ by a low-pass symbol $\tau_L(\sigma)$ (typically compactly supported or rapidly decaying), leading to a family of regularized FBP reconstructions indexed by a bandwidth parameter $L > 0$.

One may introduce a bandwidth parameter $L > 0$ and consider symbols of the form

$$\tau_L(\sigma) = |\sigma|\, w\Big(\frac{\sigma}{L}\Big),$$

where $w : \mathbb{R} \to [0, 1]$ is an even cut-off function (for instance $w \in C_c^\infty(\mathbb{R})$ with $w(\rho) = 1$ for $|\rho| \leq 1$ and $w(\rho) = 0$ for $|\rho| \geq 2$). Varying L yields a scale of reconstructions that interpolates between stronger smoothing (small L) and higher resolution (large L) [10].

Nevertheless, the cut-off function could eliminate some important frequencies. For this reason, we can consider symbols with some smooth decay to zero. Moreover, in the following section we will see that the filter can be chosen to suppress some specific frequencies.

In what follows, we consider several choices of symbols τ within the FBP framework. We use the following phantom for all the numerical tests.

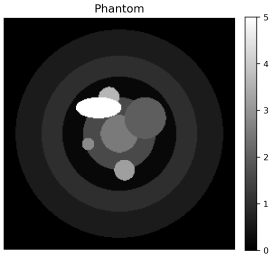


Fig. 1 Test phantom used in all numerical experiments. The image consists of concentric structures and inclusions with varying intensities, serving as a benchmark for assessing resolution and contrast in the reconstructions.

From the spectral viewpoint above, these families are naturally compared through the qualitative behavior of their symbols (growth, decay, and cut-off), which in

turn predicts the trade-off between noise suppression and resolution observed in numerical reconstructions.

It is interesting to remark that the operator $\sqrt{1+\Delta}$ can also be used as a filter. The plot of its symbol and the corresponding reconstruction are shown in Fig. 2 and Fig. 3, respectively, as item **(B)**. This operator is related to relativistic Hamiltonians, and its spectral properties are well known [12].

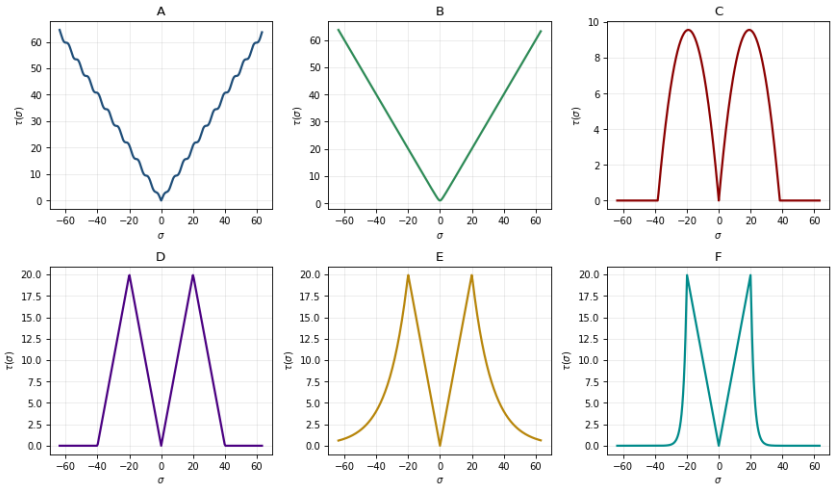


Fig. 2 Frequency-domain representation of the filters used in the FBP reconstruction. Each curve, labeled A–F, represents a symbol $\tau(\sigma)$ defining a Fourier multiplier, which determines the amplification or attenuation of frequency components.

3 Discretization and Spectral Decomposition

In this section, we use the notation presented in [11], where the image to be reconstructed is called a lattice set. The following is a general definition for the Radon transform, where $\hbar$ represents a small positive constant that scales the integer lattice.

Let $\psi : \hbar\mathbb{Z}^d \to D$, with finite support $\text{supp}(\psi) = \left\{x \in \hbar\mathbb{Z}^d : \psi(x) \neq 0\right\}$, where D is some specific set of non-negative numbers. We use $\mathscr{F}_D^d$ to denote the set of all functions with finite support in D.

Let $\psi \in \mathscr{F}_D^d$, S be a $(d-1)$-dimensional subspace of $\hbar\mathbb{Z}^d$ and $\mathscr{A}(S)$ the set of all $(d-1)$-dimensional affine subspaces in $\mathbb{R}^d$ that are parallel to S. The discrete k-dimensional X-ray of ψ parallel to S is the function $X_S\psi : \mathscr{A}(S) \to [0, \infty[$ defined by

$$X_S\psi(T) = \sum_{x \in T} \psi_x$$

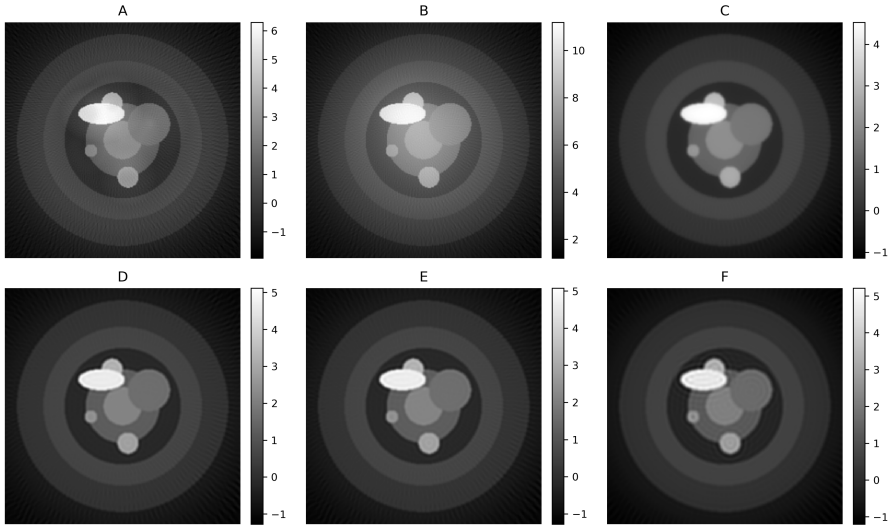


Fig. 3 Filtered back-projection reconstructions corresponding to the filters shown in Fig. 2. Each subfigure (A–F) is obtained using the symbol with the same label, illustrating how the spectral behavior of τ influences the trade-off between noise amplification and resolution.

for $T \in \mathscr{A}(S)$.

If we consider the set $\mathcal{S}_{d-1}$ of all $(d-1)$-dimensional subspaces of $\hbar\mathbb{Z}^d$, the *discrete Radon transform* is defined as

$$\begin{aligned} \mathcal{R} : \mathcal{S}_{d-1} &\to \mathbb{R}^d \\ S &\to X_S\psi(T) \end{aligned}$$

For our purposes, the dimension will be $d = 2$. Assume that $\operatorname{supp}(\psi) \subseteq \mathbb{T}^2 = \left[-\frac{1}{2}, \frac{1}{2}\right)^2$. If we discretize the domain of $\mathcal{R}\psi$ into a grid $\{\sigma_i\}_{i=-M}^{M} \times \{\varphi_i\}_{i=0}^{N-1}$, then the data measured is the matrix

$$g(i,j) = (\mathcal{R}\psi)(s_i, \varphi_j).$$

Now, we perform the Fast Fourier Transform (FFT) on each column of g and multiply the resultant matrix by the diagonal matrix M_τ defined as

$$M_\tau = \left[\tau(\sigma_{-M}), ..., \tau(\sigma_M)\right].$$

Finally, we apply the IFFT matrix and the discrete back projection formula

$$(R^* p)(x,y) = \frac{1}{N} \sum_{j=0}^{N-1} p\left(x\cos\left(\varphi_j\right) + y\sin\left(\varphi_j\right), \varphi_j\right),$$

where p is an interpolation of IFFT $\cdot\, M_\tau \cdot$ FFT(g). Note that the discrete Fourier multiplier has the decomposition IFFT$\cdot M_\tau \cdot$FFT, where the eigenvalues of it coincide with a finite subset of $\overline{\text{Ran}\,(\tau)}$ and the eigenvectors are the columns of the matrix for the FFT. The diagonal of M_τ is a finite sample of the symbol τ, then we can choose specific frequencies to amplify/reduce in order to obtain better reconstructions in a particular case of noisy data.

4 Conclusions

In the literature, it is well known that, in the presence of noisy data, the ramp filter amplifies noise. However, using a filter that attenuates high frequencies may also remove relevant information from the image.

Furthermore, the discretization process indicates that the filter becomes a vector in $\mathbb{R}^{2M+1}$ with a ramp-like shape. This allows for a more precise analysis of the frequencies that should be amplified or suppressed. Combined with machine learning techniques, this approach is expected to produce filters capable of reconstructing images affected by specific types of noise.

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